

TCP/IP Protocol Stack

Application Layer (HTTPS, FTP, SMTP, DNS)	Includes protocols like HTTP, FTP, SMTP, and DNS. These provide services and interfaces directly to end user applications
Transport Layer (TCP, UDP)	Includes TCP and UDP. Responsible for reliable or connection less data delivery, error detection and flow control
Internet Layer (IP)	Includes IP. Handles addressing, routing and packet delivery across networks
Data Link Layer (Ethernet, Wi-Fi)	Includes Ethernet, Wi-Fi. Manages physical addressing(MAC), framing and how data is transmitted over physical media.

Purpose of Each Layer

Application Layer: Enables software and users to access network resources

Transport Layer: Ensures data is delivered correctly and in order(TCP) or quickly without reliability (UDP).

Internet Layer: Routes packets across interconnected networks ensuring they reach the correct destination.

Data Link Layer: Deals with physical transmission of data, including addressing, framing, and media specific protocols.

Importance of Protocols for Internet Communication

Protocols are essential for internet communication because they establish the rules for how the devices communicate, ensuring compatibility across different hardware, software and networks. Without standardized protocols, data could not be reliably transmitted, routed, or authenticated between systems worldwide. They enable scalability, security and seamless global communication that we rely on today.